

## Hodge theory - Linear algebra

Let  $V$  be an  $\mathbb{R}$ -vector space. Let

$$V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}.$$

Then  $V_{\mathbb{C}} \cong V \oplus iV$  as a real vector space, and

$$\mathbb{C} \otimes V \oplus iV \cong$$

$$(a+ib)(v_1, v_2) = (av_1 - bv_2) + i(av_2 + bv_1).$$

We have a conjugation  $\sigma: V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$  via

$$\sigma: v \otimes \alpha \mapsto v \otimes \bar{\alpha} =: \overline{v \otimes \alpha}$$

and identify  $V = \text{Fix } \sigma \leq V_{\mathbb{C}}$ .

Conversely, let  $W$  be a  $\mathbb{C}$ -vector space with a map  $\sigma: W \rightarrow W$  such that

$$\sigma^2 = 1, \sigma(v+w) = \sigma(v) + \sigma(w), \sigma(\alpha w) = \bar{\alpha} \sigma(w).$$

Then  $W = V_{\mathbb{C}}$ , where  $V = \text{Fix } \sigma$  is a real form of  $W$ .

Let  $V, V'$  be real vector spaces. Then

$$(\text{Hom}_{\mathbb{R}}(V, V'))_{\mathbb{C}} \cong \text{Hom}_{\mathbb{C}}(V_{\mathbb{C}}, V'_{\mathbb{C}}).$$

and  $\text{Hom}_{\mathbb{C}}(V_{\mathbb{C}}, V'_{\mathbb{C}})$  gets the conjugation

$$T \mapsto \sigma' \circ T \circ \sigma.$$

Let

$$T^{a,b}(V) = \underbrace{V \otimes \dots \otimes V}_a \otimes \underbrace{V^* \otimes \dots \otimes V^*}_b,$$

$$\Lambda^r(V^*) = T^{0,r}(V) / (f \otimes f).$$

Alternatively, alternating  
multilinear maps  
 $\bigoplus_r V \rightarrow \mathbb{R}$

$$\text{Then } T^{a,b}(V_{\mathbb{C}}) \cong (T^{a,b}(V))_{\mathbb{C}}, \quad \Lambda^r(V_{\mathbb{C}}^*) \cong (\Lambda^r(V^*))_{\mathbb{C}}.$$

$$\text{Since } T^{a,b}(V) \xrightarrow{\otimes 1} (T^{a,b}(V))_{\mathbb{C}}$$

any bilinear form  $B \in T^{0,2}(V)$  gives a bilinear  
form  $B \in T^{0,2}(V_{\mathbb{C}})$  with  $\bar{B} = B$ .

Let  $W$  be a  $\mathbb{C}$ -vector space. Then the underlying  $\mathbb{R}$ -vector space  $W^{\mathbb{R}}$  comes with a map

$$J: W^{\mathbb{R}} \rightarrow W^{\mathbb{R}}, w \mapsto iw,$$

satisfying  $J^2 = -1$ . Let  $\{d_1, \dots, d_{2n}\}$  be an  $\mathbb{R}$ -basis for  $W^{\mathbb{R}}$ . Then so is

$$\{\underbrace{d_1 - Jd_{n+1}}_{e_1}, \dots, \underbrace{d_n - Jd_{2n}}_{e_n}, \underbrace{d_{n+1} + Jd_1}_{f_1}, \dots, \underbrace{d_{2n} + Jd_n}_{f_n}\}.$$

In this basis,  $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , i.e.  $Je_i = f_i$ ,  $Jf_i = -e_i$ .

Example:  $W = \mathbb{C}$ ,  $W^{\mathbb{R}} \cong \mathbb{R}^2$ ,

$\mathbb{R}$ -basis for  $W^{\mathbb{R}}$ :  $\{\operatorname{Re} z, \operatorname{Im} z\}$

$$\rightarrow \{e_1 = \operatorname{Re} z - J\operatorname{Im} z, f_1 = \operatorname{Im} z + J\operatorname{Re} z\}$$

$a - ib \qquad b + ia$

Consider the map  $J: V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$ . Since  $J^2 = -1$ ,  $V_{\mathbb{C}} = W_+ \oplus W_-$  splits into  $\pm i$ -spaces  $W_{\pm}$ . ← 4 real dimensions

Furthermore, since  $J = \overline{J}$ , for  $w \in W_{\pm}$ ,

$$J(\overline{w}) = \overline{J(w)} = \overline{\pm iw} = \mp iw,$$

$$\text{so } W_{\pm} = \overline{W_{\mp}}.$$

Note that we have an isomorphism

$$(V, J) \rightarrow W_+, v \mapsto v - iJ(v)$$

of  $\mathbb{C}$ -vector spaces (with inverse  $w \mapsto \frac{1}{2}(w + \overline{w})$ ).

An  $\mathbb{R}$ -linear map  $T: (V, \mathcal{J}) \longrightarrow (V, \mathcal{J})$  is  $\mathcal{J}$ -linear if  $T \circ \mathcal{J} = \mathcal{J} \circ T$ . In the basis above, this happens iff  $T = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$ .

The dual space  $V^*$  inherits the complex structure  $\mathcal{J}^*$ . We get the decomposition

$$V_{\mathbb{C}}^* = W_-^{\perp} \oplus W_+^{\perp},$$

where  $W_{\pm}^{\perp}$  is the  $\mp i$ -eigenspace of  $\mathcal{J}^*$ .

Note that  $W_{\pm}^{\perp} = \{ \alpha \in V_{\mathbb{C}}^* : \alpha|_{W_{\pm}} = 0 \}$ , since for  $w \in W_{\pm}$ ,  $\alpha \in W_{\pm}^{\perp}$ ,

$$\mp i \alpha(w) = (\mathcal{J}^* \alpha)(w) = \alpha(\mathcal{J}w) = \alpha(\pm i w) = \pm i \alpha(w).$$

Def: A **Hodge structure** of weight  $k$  on an  $\mathbb{R}$ -vector space  $V$  is a decomposition

$$V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$$

with  $V^{p,q} = \overline{V^{q,p}}$ .

If  $V, W$  have Hodge structures of weights  $k, l$ , resp., we give  $\text{Hom}_{\mathbb{R}}(V, W)$  a Hodge structure of weight  $k-l$  via

$$\text{Hom}_{\mathbb{C}}(V_{\mathbb{C}}, W_{\mathbb{C}})^{a,b} = \{T: T(V^{p,q}) \subseteq W^{p+a, q+b}\}.$$

In particular, if  $\mathbb{R}$  has the trivial Hodge structure of weight 0,  $V^*$  gets a Hodge structure of weight  $-k$ .

More generally,

$$(V \otimes W)^{a,b} = \bigoplus_{\substack{p+r=a \\ q+s=b}} V^{p,q} \otimes_{\mathbb{C}} W^{r,s},$$

where  $a+b = k+l$ .

Example: Let  $(V, J)$  be an  $\mathbb{R}$ -vector space with complex structure. Then  $V^*$  gets a Hodge structure via

$$(V^*)^{1,0} = W_+^{\perp}, \quad (V^*)^{0,1} = W_-^{\perp},$$

which induces a Hodge structure on  $\wedge^r(V^*)$  of weight  $r$  via

$$\wedge^{p,q}(V^*) = \underbrace{(V^*)^{1,0} \wedge \dots \wedge (V^*)^{1,0}}_p \wedge \underbrace{(V^*)^{0,1} \wedge \dots \wedge (V^*)^{0,1}}_q$$

Def (alternative): A Hodge Structure of weight  $k$  on an  $\mathbb{R}$ -vector space  $V$  is a filtration

$$V_{\mathbb{C}} \supseteq \dots \supseteq F^p \supseteq F^{p-1} \supseteq \dots$$

such that

$$V_{\mathbb{C}} = F^p \oplus \overline{F^{k-p+1}}$$

Note that this definition is equivalent to the first one via

$$F^p = \bigoplus_{a \geq p} V^{a, k-a}, \quad V^{p,q} = F^p \cap \overline{F^q} \quad (p+q=k).$$

Let  $(V, J)$  be an  $\mathbb{R}$ -vector space with complex structure. Let  $B: V \times V \rightarrow \mathbb{R}$  be a symmetric bilinear form with  $B(Jv, Jw) = B(v, w)$ . Define

$$\omega: V_{\mathbb{C}} \times V_{\mathbb{C}} \rightarrow \mathbb{C}, \quad \omega(u, v) = B(Ju, v).$$

Then

- $\omega \in \Lambda^2(V_{\mathbb{C}}^*)$

$$\omega(u, v) = B(Ju, v) = B(J^2 u, Jv) = -B(u, Jv) = -\omega(v, u)$$

- $\omega$  is real (i.e.  $\bar{\omega} = \omega$ )  $\leadsto \omega \in \Lambda^2(V^*)$

$$\omega(\bar{u}, \bar{v}) = B(J\bar{u}, \bar{v}) \stackrel{J^2 = -\text{id}}{=} B(\bar{J}u, \bar{J}v) \stackrel{B \text{ real}}{=} \overline{B(Ju, v)} = \overline{\omega(u, v)} = \omega(u, v)$$

- $\omega \in \Lambda^{1,1}(V^*)$

$$\text{This follows since } \omega(J \cdot, J \cdot) = \omega(\cdot, \cdot)$$

(Conversely, if  $\omega \in \Lambda^{1,1}(V^*) \cap \Lambda^2(V^*)$  then

$$B(u, v) := \omega(u, Jv)$$

is a symmetric real bilinear form, compatible with  $J$ .

Thm: Let  $(V, J)$  be an  $\mathbb{R}$ -vector space with complex structure. The following data are equivalent:

$$H(v, w) = \overline{H(w, v)}$$

- A Hermitian form  $H$  on the  $\mathbb{C}$  vector space  $(V, J)$ ;
- A symmetric bilinear form  $B$  on  $V$ , compatible with  $J$ ;
- An element  $\omega \in \Lambda^{1,1}(V^*) \cap \Lambda^2(V^*)$ .

Proof:

|       |                                                            |
|-------|------------------------------------------------------------|
| given |                                                            |
| H     | $B = \operatorname{Re} H, \omega = -\operatorname{Im} H$   |
| B     | $\omega(\cdot, \cdot) = B(J\cdot, \cdot), H = B - i\omega$ |
| ω     | $B(\cdot, \cdot) = \omega(\cdot, J\cdot), H = B - i\omega$ |

We will show the first line. Some of the remaining details were already shown.

Since  $H$  is Hermitian,

$$B(v, w) - i\omega(v, w) = B(w, v) + i\omega(w, v),$$

so  $B$  is symmetric,  $\omega$  alternating. Furthermore, if  $z_1, \dots, z_n$  is a basis for  $W^{1,0}$ ,  $\bar{z}_1, \dots, \bar{z}_n$  the conjugate basis for  $W^{0,1} = \overline{W^{1,0}}$ , then

$$H = \sum h_{ij} z_i \otimes \bar{z}_j$$

where  $(h_{ij})$  is Hermitian (its own conjugate transpose). Hence,

$$\begin{aligned} \omega = H - \bar{H} &= \sum h_{ij} z_i \otimes \bar{z}_j - \overline{\sum h_{ij} z_i \otimes \bar{z}_j} \\ &= \sum h_{ij} z_i \otimes \bar{z}_j - h_{ji} \bar{z}_i \otimes z_j \quad (\bar{H} = H^*) \\ &= \sum h_{ij} z_i \otimes \bar{z}_j - h_{ij} \bar{z}_j \otimes z_i \\ &= \sum h_{ij} z_i \wedge \bar{z}_j. \end{aligned}$$

Since  $H$  is Hermitian,  $H(Jv, Jw) = H(v, w)$ , so the same is true about  $B$ .





## Hodge theory on a compact Kähler manifold

Let  $X$  be an <sup>oriented</sup> compact Riemannian manifold. We define the  $L^2$ -metric for  $\psi, \eta \in \mathcal{E}^i(X)$  <sup>differential forms</sup> by

$$(\psi, \eta)_{L^2} = \int_X g(\psi, \eta) d\text{Vol}.$$

We define the Hodge star operator  $*$ :  $\mathcal{E}^i \longrightarrow \mathcal{E}^{n-i}$  by requiring that

$$g(\psi, \eta) d\text{Vol} = \psi \wedge * \eta.$$

One checks (in coordinates) that  $*^2 = (-1)^{i(n-i)}$ . We define the adjoint

$$\partial^* = (-1)^{i+1} * \circ \partial \circ * = (-1)^{i+1} *^{-1} \partial *.$$

Then

$$(\partial \psi, \eta) = (\psi, \partial^* \eta)$$

for  $\psi \in \mathcal{E}^i, \eta \in \mathcal{E}^{i+1}$ .

We define the Laplacian

$$\Delta = \partial^* \partial + \partial \partial^*.$$

We call the forms in

$$\mathcal{H}^i(X) = \mathcal{E}^i(X) \cap \text{Ker } \Delta$$

harmonic  $i$ -forms.

Theorem:  $\mathcal{H}^i(X) \xrightarrow{\cong} H^i(X; \mathbb{R})$ .

ARE THE SIGNS RIGHT?

Now let  $X$  be a complex manifold. Then the tangent space at each point can be complexified into

$$T_{\mathbb{C}} = T^{1,0} \oplus T^{0,1},$$

where

$$T^{1,0} = \{u - i\mathcal{I}u : u \in T\}, \quad T^{0,1} = \overline{T^{1,0}}.$$

see page 3, this is the  $+i$ -space of  $\mathcal{I}$

Since the changes of coordinates are holomorphic,  $\mathcal{I}$  is defined globally via  $\mathcal{I} \frac{\partial}{\partial x} = \frac{\partial}{\partial y}$ , and we let  $\frac{\partial}{\partial z} = \frac{1}{2}(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y})$ .

As before, we get a splitting

$$\Sigma^k \otimes \mathbb{C} = \bigoplus_{p+q=k} \Sigma^{p,q}.$$

Note that  $\mathcal{J}: \Sigma^{p,q} \longrightarrow \Sigma^{p+1,q} \oplus \Sigma^{p,q+1}$ . We define  $\partial = p_1 \circ \mathcal{J}$ ,  $\bar{\partial} = p_2 \circ \mathcal{J}$ :

$$\mathcal{J}: \Sigma^{p,q} \longrightarrow \Sigma^{p+1,q} \oplus \Sigma^{p,q+1}$$

$$\begin{array}{c} p_1 \\ \swarrow \\ \Sigma^{p+1,q} \end{array}$$

$$\begin{array}{c} p_2 \\ \searrow \\ \Sigma^{p,q+1} \end{array}$$

Now let  $X$  have a Hermitian metric,  $E$  a Hermitian vector bundle.  
We define the conjugate linear operator

$$\bar{*}: \Sigma^{p,q} \longrightarrow \Sigma^{n-q, n-p}, \quad \alpha \longmapsto \bar{*}\alpha.$$

Let  $\iota: E \xrightarrow{\cong} E^\vee$ . We define

$$\bar{*}_E: \Sigma^{p,q}(E) \longrightarrow \Sigma^{n-q, n-p}(E^\vee), \quad \alpha \otimes e \longmapsto \bar{*}\alpha \otimes \iota(e).$$

We get an inner product of  $E$ -valued forms by

$$(\alpha, \beta) = \int \alpha \wedge \bar{*}_E \beta,$$

We define an operator  $\bar{\partial}_E: \Sigma^{p,q}(E) \longrightarrow \Sigma^{p,q+1}(E)$  by  $\alpha \otimes e \longmapsto \bar{\partial}\alpha \otimes e$ . We denote its cohomology by  $H_{\bar{\partial}}^{p,q}(E)$ . Similar to before, we define its formal adjoint and the corresponding Laplacian.

Theorem:  $\text{Har}^{p,q}(E) \xrightarrow{\cong} H_{\bar{\partial}}^{p,q}(E)$ .

Note that  $\bar{*}_E$  commutes with  $\Delta_E$ , so that it preserves harmonic forms.

In particular,

(Corollary (Serre Duality)):  $\bar{*}_E: H^q(X, \Omega_X^p(E)) \xrightarrow{\cong} H^{n-q}(X, \Omega_X^{n-p}(E^\vee))^\vee$ .

We call the Hermitian metric  $h$  Kähler if its 2-form, the Kähler form, is closed. It can be shown that a Hermitian metric is Kähler iff  $\mathcal{L}$  is parallel.

Theorem: Let  $X$  be compact Kähler. Then

$$H_{\mathbb{R}}^m(X) \otimes \mathbb{C} = \bigoplus_{p+q=m} H^{p,q}(X)$$

is a Hodge structure for  $H_{\mathbb{R}}^m(X)$ .

# Hodge structure as $SL(2; \mathbb{R})$ representation

Let  $sl(2; \mathbb{R})$  have basis

$$e = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$[e, \lambda] = b, \quad [e, b] = -2e, \quad [\lambda, b] = 2\lambda.$$

Let  $\rho: sl(2; \mathbb{R}) \rightarrow \text{End } V$  be a real representation. Let

$$L = \rho(e), \quad \Lambda = \rho(\lambda), \quad B = \rho(b).$$

We let  $\overset{\text{degree } +2}{\nearrow} \quad \overset{\text{degree } -2}{\nwarrow}$

$$V_{\text{prim}} = \text{Ker } \Lambda.$$

It can be shown that all representations of  $sl(2; \mathbb{R})$  are completely reducible and that they share many properties with the standard representations on the homogeneous polynomials of degree  $d$ ,  $V_d$ , via

$$L = -y \frac{\partial}{\partial x}, \quad \Lambda = -x \frac{\partial}{\partial y}, \quad B = [L, \Lambda] = y \frac{\partial}{\partial y} - x \frac{\partial}{\partial x},$$

with

$$V_{\text{prim}} = S_p(x^d).$$

Lemma: Let  $sl(2; \mathbb{R}) \otimes V$ . Then

- $L, \Lambda$  are nilpotent,  $B$  diagonalisable with spaces  $V^m$  and  $L V^m \subset V^{m+2}, \Lambda V^m \subset V^{m-2}$ .
- $V_{\text{prim}} \cap V^m \neq 0$  some  $m$ ; if  $w \in V_{\text{prim}} \cap V^m, L^{m+1} w = 0 \neq L^m w$  then  $w \in V_{-m}$ .
- if  $V_{\text{prim}}^m = V_{\text{prim}} \cap V^m$  then

$$V_{\text{prim}} = \bigoplus_{m \in \mathbb{N}} V_{\text{prim}}^{-m}$$

and

$$L^r: V_{\text{prim}}^{-m} \hookrightarrow V^{-m+2r} \quad (r \leq m),$$

$$L^r(V_{\text{prim}}^{-m}) = 0 \quad (r > m).$$

.

$$V = \bigoplus_{r \in \mathbb{N}} L^r V_{\text{prim}},$$

$$V = \bigoplus_{r \in \mathbb{N}_n} L^r(V_{\text{prim}}^{m-2r}), \text{ where } L^r: V_{\text{prim}}^{m-2r} \xrightarrow{\cong} L^r(V_{\text{prim}}^{m-2r}).$$

Now let  $X$  be a compact Kähler manifold again. Let  $\omega_h$  denote its Kähler class. Let

$$L: \Sigma^i \longrightarrow \Sigma^{i+2}, \quad \alpha \longmapsto \omega_h \wedge \alpha,$$

and let

$$\Lambda = L^* = *^{-1} L *$$

It can be shown that

$$[L, \Lambda]u = (m - u)u$$

for  $u \in \Lambda^m T^*X$ . This defines an action of  $sl(2; \mathbb{R})$  on harmonic forms, and hence on cohomology. We let

$$\begin{aligned} H_{\text{prim}}^m(X) &= \text{Ker}(\Lambda: H^m(X; \mathbb{C}) \longrightarrow H^{m-2}(X; \mathbb{C})), \\ &= \text{Ker}(L^{n-m+1}: H^m(X; \mathbb{C}) \longrightarrow H^{2n-m+2}(X; \mathbb{C})). \end{aligned}$$

In particular, primitive cohomology does not depend on the metric.

We get decompositions

$$H_{\text{prim}}^m(X) = \bigoplus_{p+q=m} H_{\text{prim}}^{p,q}(X),$$

$$H^m(X; \mathbb{C}) = \bigoplus_{r \geq (m-n), 0}^r H_{\text{prim}}^{m-2r}(X; \mathbb{C}),$$

$$H^{p,q}(X) = \bigoplus_{r \geq p+q-n, 0}^r H_{\text{prim}}^{p-r, q-r}(X).$$

Theorem (Hard Lefschetz):

$$L^{n-k}: H^k(X; \mathbb{C}) \xrightarrow{\cong} H^{2n-k}(X; \mathbb{C}), \quad (k \leq n)$$

$$L^{n-(p+q)}: H^{p,q}(X) \xrightarrow{\cong} H^{n-q, n-p}(X). \quad (p+q \leq n)$$

We call the action of the complex structure  $J$  on  $H_{\mathbb{R}}^*(X; \mathbb{C})$  the **Weil operator**  $L$ , which acts as

$$L|_{H^{p,q}(X)} = i^{p-q}.$$

Theorem (Hodge-Riemann bilinear relations): For  $k \in \mathbb{Z}$ , let  $\varepsilon(k) = (-1)^{\frac{1}{2}k(k-1)}$ . Define

$$Q: H_{\mathbb{R}}^k(X; \mathbb{C}) \times H_{\mathbb{R}}^k(X; \mathbb{C}) \longrightarrow \mathbb{C},$$

$$Q(\alpha, \beta) = \varepsilon(k) \int \alpha \wedge \beta \wedge \omega^{n-k}.$$

Then  $Q$  is  $(-1)^k$ -symmetric <sup>(symmetric for  $k$  even)</sup><sub>(skew-symmetric for  $k$  odd)</sub>, and

- $Q(H^{p,q}, H^{r,s}) = 0$  if  $(r,s) \neq (q,p)$ ,
- $u \in H_{\text{prim}}^{p,q} \setminus 0 \Rightarrow i^{p-q} Q(u, \bar{u}) = (u, u) > 0$ .

Proof: Only the last point is non-trivial. It follows from  $\varepsilon(k) i^{p-q} \omega^{n-k} \wedge \bar{u} = (n-k)! \cdot \bar{u}$ .

A basic Hodge structure of weight  $-2m$  are the Hodge structures of Tate

$$\mathbb{Z}(m) = (\mathbb{Z}\pi i)^m \mathbb{Z} \subset \mathbb{C}, \quad \mathbb{Z}(m) \otimes \mathbb{C} = (\mathbb{Z}(m) \otimes \mathbb{C})^{-m, -m}$$

and for any  $V$  with HS of weight  $k$ ,

$$V(m) = V \otimes_{\mathbb{Z}} \mathbb{Z}(m)$$

with HS of weight  $k-2m$ .

If  $X$  is a compact complex manifold then

$$\text{Tr}: H^{2n}(X; \mathbb{R}) \xrightarrow{\cong} \mathbb{R}(-n), \quad \omega \mapsto \left(\frac{1}{2\pi i}\right)^n \int_X \omega$$

is an isomorphism of Hodge structures.

Note: Above, we gave  $\mathbb{Z}$  a Hodge structure. In the same way, we can define a Hodge structure for any  $R$ -module, where  $R \subset \mathbb{R}$  is a subring.

In this way, the cohomology of a compact Kähler manifold actually carries a  $\mathbb{Z}$ -Hodge structure.



Definition: A **polarization** of an  $\mathbb{R}$ -Hodge structure  $V$  of weight  $k$  is a bilinear form

$$Q: V \otimes V \longrightarrow \mathbb{R}$$

which is  $(-1)^k$ -symmetric and such that

- $(F^m)^\perp = F^{k-m+1}$ ,

- the Hermitian form on  $V_{\mathbb{C}}$  given by  $Q(u, \bar{v})$  is pos. def.

Weil operator:

$i^{p-q}$  on  $V^{p,q}$

